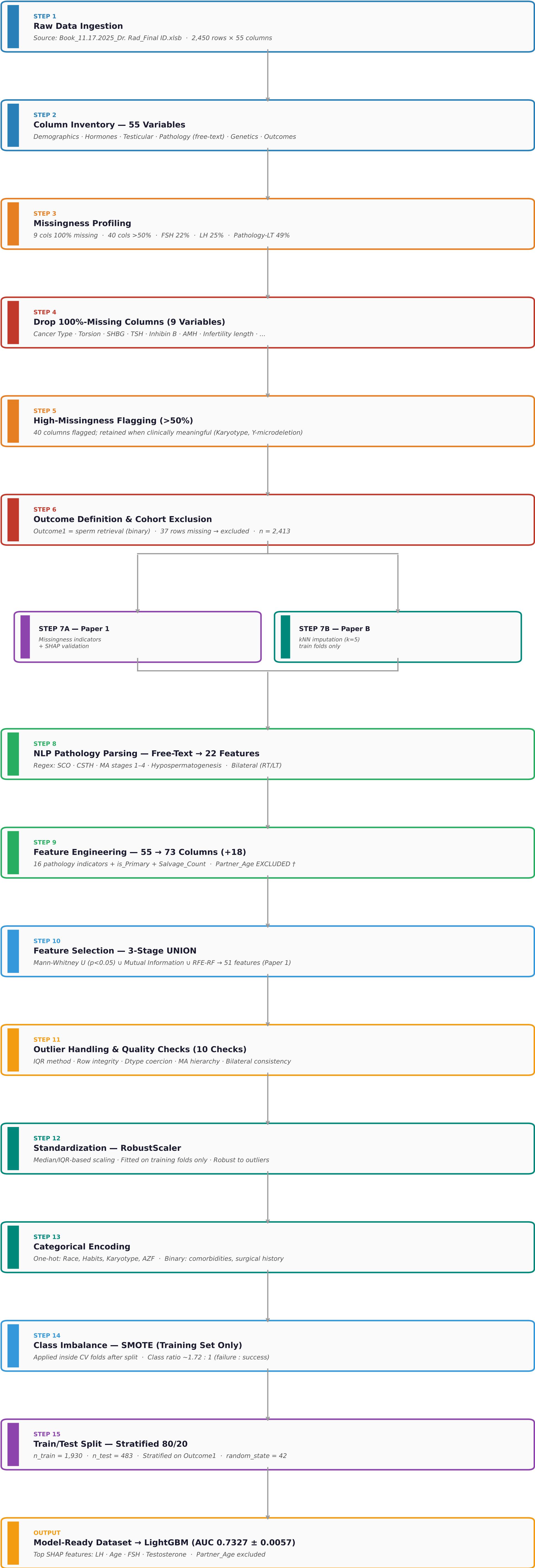


Data Cleaning & Preprocessing Pipeline

NOA Micro-TESE ML Prediction — Royan Institute Cohort (n = 2,413)



Legend:

Data Ingestion

Missingness Analysis

Exclusion/Filtering

Engineering/NLP

Feature Selection

Preprocessing

Quality/Output

Split/Branch